MINI-PROJECT REPORT

Survey Report on Types of Blockchains and Their Real-Time Use Cases

Subject: Blockchain Technology

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Objective of the Assignment

To learn about the different use cases and real-time applications of Blockchain, and to perform a survey on one of the case studies.

1. Introduction to Blockchain

Blockchain is a distributed, decentralized and immutable digital ledger that records transactions across a network of computers (called nodes). Instead of a single central authority such as a bank or government maintaining the records, every participant holds a copy of the ledger, and new entries are added only when the network agrees on their validity. The concept was introduced in 2008 by the pseudonymous Satoshi Nakamoto as the foundation of Bitcoin, but its use has since expanded far beyond cryptocurrency.

Transactions are grouped into **blocks**. Each block contains a list of transactions, a timestamp, a nonce and the cryptographic hash of the previous block. Because every block points to the hash of the one before it, the blocks form a chain; altering any past block changes its hash and breaks all subsequent links, which makes tampering easy to detect.

1.1 Key Characteristics

- **Decentralization:** no single point of control or failure; the ledger is shared across many nodes.
- **Immutability:** once a block is confirmed, its data is practically impossible to change without network consensus.
- **Transparency and traceability:** participants can verify the history of any recorded asset or transaction.
- **Security:** cryptographic hashing and digital signatures protect integrity and authenticity.
- **Consensus:** nodes agree on the ledger state using mechanisms such as Proof of Work, Proof of Stake or PBFT.
- **Smart contracts:** self-executing programs stored on the blockchain that enforce agreed rules automatically.

1.2 How a Transaction Works

- A user initiates a transaction and signs it with their private key.
- The transaction is broadcast to the peer-to-peer network of nodes.
- Validators verify the signature and check the transaction against the ledger rules.
- Verified transactions are bundled into a new block and the network reaches consensus.
- The block is appended to the chain and the updated ledger is replicated across all nodes.

2. Types of Blockchain

Blockchains are classified mainly by who is allowed to join the network and who can validate transactions. There are four common types.

2.1 Public Blockchain

A public blockchain is permissionless: anyone can join, read the ledger, submit transactions and take part in consensus. It is fully decentralized and highly transparent, but typically has lower throughput and higher energy or fee costs. *Examples:* Bitcoin, Ethereum, Solana.

2.2 Private Blockchain

A private blockchain is controlled by a single organization that decides who can participate, read and write. It offers higher speed, better privacy and easier governance, but is more centralized and requires trust in the owner. *Examples:* internal enterprise ledgers, Hyperledger Fabric deployments within one company.

2.3 Consortium (Federated) Blockchain

A consortium blockchain is governed by a group of pre-selected organizations rather than one entity. Validators are the consortium members, so it balances decentralization with efficiency and privacy. *Examples:* R3 Corda (banking), IBM Food Trust, Energy Web Chain.

2.4 Hybrid Blockchain

A hybrid blockchain combines features of public and private chains. Some data stays private and permissioned while selected records can be verified publicly, giving organizations flexibility and controlled transparency. *Example:* Dragonchain, XinFin-style hybrid networks.

2.5 Comparison of Blockchain Types

Feature	**Public**	**Private**	**Consortium**	**Hybrid**
Access	Open to everyone	Single organization decides	Group of organizations	Mix of open and restricted
Decentralization	High	Low	Partial	Partial
Speed / Scalability	Low to moderate	High	High	High
Privacy	Pseudonymous, public data	High	High	Configurable
Consensus	PoW, PoS	Voting / approved validators	PBFT, Raft, PoA	Varies by layer
Energy cost	Can be high (PoW)	Low	Low	Low to moderate
Examples	Bitcoin, Ethereum	Enterprise Hyperledger Fabric	Corda, IBM Food Trust	Dragonchain

3. Different Applications of Blockchain

Blockchain is used wherever multiple parties need a shared, trustworthy record without depending on a single intermediary. The major real-time application areas are listed below.

Domain	Application	Real-world example
Finance and Banking	Cryptocurrencies, cross-border payments, trade finance, tokenized assets	Bitcoin, Ripple (XRP), JPMorgan Onyx
Supply Chain	Product tracking, provenance and food safety traceability	IBM Food Trust (Walmart), Maersk TradeLens
Healthcare	Secure sharing of patient records, drug authenticity tracking	MediLedger, Medicalchain
Government and Voting	Digital identity, land registry, tamper-resistant e-voting	Estonia e-Residency and KSI, Georgia land registry
Education	Verifiable academic certificates and credentials	MIT Blockcerts
Energy	Peer-to-peer energy trading and renewable energy certificates	Power Ledger, Energy Web
Real Estate	Property records, tokenized ownership, smart-contract based sale	Propy
Insurance	Automated claims through smart contracts, fraud reduction	Etherisc, AXA (Fizzy)
Media and Entertainment	Digital rights management, NFTs, royalty distribution	OpenSea, Audius
Cybersecurity and Digital Forensics	Tamper-proof logs and chain of custody for digital evidence	Blockchain-based evidence ledgers

4. Detailed Survey on One Use Case: Blockchain in Supply Chain (Food Traceability)

4.1 Background and Problem Statement

Modern food supply chains involve farms, processors, distributors, logistics firms, retailers and regulators spread across many countries. Records are maintained in separate databases or on paper, which leads to poor visibility, delays and a lack of trust between parties. During a foodborne illness outbreak, tracing contaminated produce back to its source with traditional methods can take days or even weeks, resulting in large recalls, financial losses, wasted food and risk to public health. Counterfeiting, mislabeling and fraud (for example, falsely marketing products as organic) further reduce consumer confidence.

4.2 Why Blockchain is Suitable

- Multiple untrusting parties need to share a common, consistent version of the truth.
- An immutable audit trail prevents records of origin, handling and storage from being altered after the fact.
- Permissioned access lets companies share data selectively while protecting commercial secrets.
- Smart contracts can automate compliance checks, payments and quality alerts.

4.3 Case Study: IBM Food Trust and Walmart

IBM Food Trust is a consortium blockchain solution built on Hyperledger Fabric, a permissioned enterprise blockchain framework hosted by the Linux Foundation. It connects growers, suppliers, processors, distributors and retailers on a shared network. Walmart was one of the earliest and most prominent adopters, running pilots on products such as sliced mangoes in the United States and pork in China, and later requiring many of its leafy green suppliers to join the platform.

The widely reported Walmart pilot showed that tracing the origin of a package of mangoes, which took about seven days using conventional paperwork and phone calls, could be reduced to a couple of seconds by querying the blockchain record. This demonstrates the key benefit of the approach: fast and precise identification of affected batches, so that only the contaminated products are removed instead of an entire category.

4.4 System Architecture and Working

- **Data capture:** at each stage (harvest, processing, packing, shipping, store arrival) participants record events such as batch ID, location, date, temperature and certifications using scanners, IoT sensors or manual entry.
- **Identification:** products are tagged with barcodes, QR codes or RFID so physical items map to digital records.
- **Permissioned ledger:** events are written as transactions to a Hyperledger Fabric network; only authorized members can read or write specific data channels.
- **Consensus and validation:** endorsing peers validate each transaction before it is ordered and committed to the ledger.
- **Smart contracts (chaincode):** enforce rules, for example flagging a shipment when the recorded temperature exceeds a threshold.
- **Access and query:** retailers, regulators and, in some cases, consumers can scan a code to view a product's journey from farm to shelf.

4.5 Benefits Observed

Aspect	Traditional system	Blockchain-based system
Traceability time	Days to weeks	Seconds to minutes

Aspect	Traditional system	Blockchain-based system
Data integrity	Records can be altered or lost	Immutable, tamper-evident records
Trust between parties	Depends on intermediaries	Shared, verifiable ledger
Recall precision	Broad recalls, high waste	Targeted removal of affected batches
Transparency for consumers	Limited	Provenance visible through QR scan
Auditing and compliance	Manual and slow	Automated and continuous

4.6 Challenges and Limitations

- **Garbage in, garbage out:** blockchain guarantees that recorded data is not changed, but cannot guarantee the data was correct when entered.
- **Adoption cost and effort:** small farmers and suppliers may lack the technology or training needed to participate.
- **Interoperability:** competing platforms and standards make it hard to connect different networks.
- **Scalability and performance:** large volumes of item-level data require careful design.
- **Governance and data privacy:** deciding who owns data and who can see it needs clear agreements and regulation compliance.

4.7 Comparison with Other Supply Chain Blockchains

Platform	Type	Focus
IBM Food Trust	Consortium (Hyperledger Fabric)	Food safety and traceability
Maersk and IBM TradeLens	Consortium (Hyperledger Fabric)	Global shipping and trade documents (platform was discontinued in 2023)
VeChain	Public, enterprise-oriented	Product authenticity and luxury goods tracking
OriginTrail	Public decentralized knowledge graph	Data exchange and verification across supply chains

4.8 Findings of the Survey

The survey shows that consortium (permissioned) blockchains are the most practical choice for enterprise supply chains because they combine shared trust with privacy, speed and controlled governance, whereas fully public chains are less suitable for confidential commercial data. Blockchain delivers the greatest value when combined with IoT devices and good data-entry practices, and when a critical mass of participants joins the network. The failure of some large platforms, such as TradeLens, indicates that business collaboration and governance are as

important as the technology itself.

5. Conclusion

Blockchain is a powerful technology for creating trusted, transparent and tamper-resistant records among parties that do not fully trust each other. Public, private, consortium and hybrid blockchains each offer different trade-offs among decentralization, performance and privacy, so the right type must be chosen according to the application. From finance and healthcare to education, energy and government services, blockchain is being applied in many real-time scenarios. The detailed study of food traceability through IBM Food Trust and Walmart shows how a consortium blockchain can cut traceability time from days to seconds, improve food safety and increase consumer trust, while also highlighting practical challenges such as data quality, adoption and governance.

References

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